

Related Work Data Fusion

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1 Regression Models and Life Tables

We consider a population of individuals; for each individual, we observe either the time to “failure” or the time to “loss” or censoring. For censored individuals, we know only that the time to failure is greater than the censoring time.

Let T be a random variable representing failure time. Let $F(t) = \mathbb{P}[T \geq t]$. Let $\lambda(t)$ be the hazard, or age-specific failure rate, i.e.

$$\lambda(t) = \lim_{\Delta \rightarrow 0^+} \frac{\mathbb{P}[t \leq T \leq t + \Delta \mid T \geq t]}{\Delta}.$$

The probability that failure happens at time t , given that it has not yet occurred. If T is discrete, then

$$\lambda(t) = \sum \lambda_{u_j} \delta(t - u_j),$$

where $\lambda_t = \mathbb{P}[T = t \mid T \geq t]$.

Let n individuals be observed to failure and the rest be censored. The only information about the failure time of a censored individual is that it exceeds the censoring time. The failure times are given by

$$t_{(1)} < t_{(2)} < \dots < t_{(n)}.$$

Let $m_{(i)}$ be the number of units with failure time equal to $t_{(i)}$. The risk set $\mathcal{R}(t)$ is the set of individuals at risk at time t . Let $r_{(i)}$ be the number of such individuals for $t = t_{(i)}$. The product limit estimate of the distribution is obtained by taking estimated conditional probabilities that agree with the observed distributional frequencies:

$$\hat{\lambda}(t) = \sum_{i=1}^k \frac{m_{(i)}}{r_{(i)}} \cdot \delta(t - t_{(i)}).$$

MLE estimate in the family of all possible distributions.

Suppose we also observe measurements x (covariates) for each unit in addition to failure time. Goal is to assess relationship between failure times and x . Posit a model where the hazard is

$$\lambda(t; x) = \exp(z_{(i)}\beta) \cdot \lambda_0(t).$$

The probability of failure at time $t_{(i)}$ is given by

$$\exp(z_{(i)}\beta) \Big/ \sum_{l \in \mathcal{R}(t_{(i)})} \exp(z_{(l)}\beta).$$

Our stationarity assumption implies that time-dependent factors have only a multiplicative impact on the probability of response.